The Benchmark-to-User Generalization Gap in Agentic LLM Search Behavior

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Paper under double-blind review at ICLR 2026

Abstract

LLM agents equipped with web-search tools are increasingly evaluated on benchmark-style multi-hop questions, and their tool-use behavior is reported as if it characterized how they would act in deployment. We show, on three frontier agentic systems (Gemini 3.1 Pro, Qwen 3.5 122B, Nemotron Nano 30B), that this inference does not hold: *agentic behavior does not transfer from benchmark-style questions to user-like questions.* We use a hop-level diagnostic protocol — per-hop semantic entropy (Farquhar et al., 2024) on closed-book samples, combined with LLM-judge attribution of each emitted search query back to a specific reasoning hop — to measure, on every question, the volume and per-hop target of the agent's search calls and how those calls relate to the agent's own parametric uncertainty. On 600 staleness-filtered MuSiQue (Trivedi et al., 2022) questions, rewriting the benchmark questions into natural, user-styled prose that preserves the underlying information need but removes the explicit hop chain collapses search volume by factors of 1.8–4.2× and redirects the surviving searches *away from* the hops the same model had flagged as uncertain — the dominant cell-level shift is Necessary search → Truly missed. Two probes identify the mechanism: a commitment-locus analysis shows the model fixing on a per-hop answer *before* issuing any search for that hop (e.g. pre-search commitment on uncertain hops for Gemini rises from 23.5% to 38.1%), and a cascade decomposition rules out the alternative explanation that the missed-hop expansion is downstream cascade failure from earlier wrong hops. On real user queries from a curated ShareChat slice the direction of effect reproduces: search-tool invocation drops sharply across all three models. Finally, we show the gap is closable: an LoRA fine-tune of Nemotron on a mix of phrasing-augmented and uncertainty-aware on-policy rollouts cuts the Missed cell on the user-style distribution by 8.9 percentage points while raising mean searches per question by 76% — a real but Pareto-bounded improvement. Together, these results identify a benchmark-to-user generalization gap that current agentic evaluations do not measure, isolate the mechanism that produces it, and demonstrate a first training-time fix.

1 Introduction

Contributions. We claim four contributions, in order of how load-bearing they are for the paper:

1. **A mechanism for the benchmark-to-user gap, not just an observation of it.** Two probes — pre-search commitment locus and missed-cell cascade decomposition — jointly show that the per-hop search-omission expansion under user-like phrasing is *upstream* of retrieval (the agent commits to an answer before any search query attributable to that hop) rather than downstream cascade failure from earlier wrong hops. To our knowledge, no prior framing-sensitivity work (Jang et al., 2026; Cheng et al., 2026; Rabinovich et al., 2025) isolates this mechanism.

2. **A first training-time fix.** An LoRA fine-tune of Nemotron Nano 30B on phrasing-augmented and uncertainty-aware on-policy rollouts partially closes the gap on the same model's user-style distribution (Missed cell −8.9 pp, Effective +8.9 pp), at the cost of a 76% rise in mean searches per question. This is, to our knowledge, the first quantitative demonstration that this particular gap is closable by fine-tuning.

3. **A hop-level diagnostic instrument** (per-hop semantic entropy × LLM-judge query-to-hop attribution) that is invariant to question phrasing and so makes "did the behavior transfer?" a directly measurable question.

4. **Naturalistic-query corroboration** on a curated ShareChat slice of real user queries, showing the search-volume drop is not an artifact of synthetic paraphrase rewriting.

2 Related Work

Agentic tool-use evaluation and its robustness. Tool-use benchmarks evaluate end-task accuracy or coarse whether-to-use-a-tool correctness at the question level: WTU-Eval (Ning et al., 2024) is canonical, and the broader agentic stack (Schick et al., 2023; Wang et al., 2025; Asai et al., 2024) is evaluated similarly. Step-level retrieval gating has been studied in pipeline-RAG settings — FLARE (Jiang et al., 2023), DeepRAG (Guan et al., 2025), HiPRAG (Wu et al., 2025a), ReaLM-Retrieve (Guo et al., 2026) — and SMART-ER (Qian et al., 2025) and Wu et al. (2025b) introduce over- and under-search rates as calibration metrics. These works share an implicit assumption: that behavior measured on a benchmark characterizes how the same agent would act on the same information need expressed in arbitrary surface forms. *Our work targets this assumption directly*, asking whether the agentic behavior reported on benchmark-style multi-hop questions generalizes to user-like rewrites of the same information need and to natural conversational queries.

Parametric vs. contextual knowledge. A growing line studies how LLMs balance internal priors against retrieved context (Cheng et al., 2024a; Wu et al., 2024a; Farahani and Johansson, 2024; Kim et al., 2025; Wadhwa et al., 2026; Tao et al., 2024; Zhang et al., 2024). Recent work moves toward per-token conflict arbitration: AdaCAD (Adacad et al., 2025), CoCoA (Khandelwal et al., 2025), Micro-Act (Huo et al., 2025). These study *conflict resolution* after retrieval has been triggered. We use this literature as motivation: if integrating retrieved context is fragile downstream, then the upstream decision of *whether to retrieve at all* on a specific hop is itself a load-bearing calibration problem — and one whose behavior should be measured under the surface-form variation real users impose.

Uncertainty quantification and uncertainty-aware retrieval. We use semantic entropy (Farquhar et al., 2024; Kunitomo-Jacquin et al., 2026) as our operational definition of "knows / does not know" at hop granularity, with Semantic Entropy Probes (Kossen et al., 2024) as a cheap single-pass alternative. The closest training-time analogues are SEARAG (SEARAG, 2025) (semantic-entropy-gated pipeline RAG), SUGAR (Zubkova et al., 2025), TARG (Wang et al., 2026), SeaKR (Yao et al., 2025) and Wu et al. (2025b) (uncertainty-driven adaptive retrieval). These works *use* uncertainty to drive a retrieval gate; we use it as a *diagnostic* reference to measure whether deployed agentic systems' search decisions track their own parametric uncertainty, and whether this tracking survives a distribution shift in how the question is phrased. Wang et al. (2025b) formalize this as the Decision-Knowledge Alignment Principle; we provide its first naturalistic-query audit.

Framing and paraphrase sensitivity — the closest prior art. A small but very recent literature documents framing-induced flips in tool-use behavior. Jang et al. (2026) measure retrieval-decision flip rates of up to 55% under meaning-preserving paraphrases in adaptive RAG; Cheng et al. (2026) formalize the "knowing-doing gap" — a mismatch between a model's internal recognition that a tool is needed and its actual generation of tool-call tokens, amplified under conversational framing; Rabinovich et al. (2025) document output-mode collapse in function-calling under prompt rephrasing. These works *observe* the gap at the question level. Our two load-bearing differentiators are downstream of that observation: (a) we identify the *mechanism* — the locus of the per-hop answer moves to *before* any search query is issued, and the missed-cell expansion is genuine missing rather than cascade failure from a wrong upstream hop (§4.3); and (b) we show the gap is *closable* by phrasing-augmented, uncertainty-aware LoRA fine-tuning on on-policy rollouts (§4.5). A hop-level diagnostic instrument and naturalistic-query corroboration are supporting contributions. Compositional brittleness under surface variation in non-agentic multi-hop settings is documented by Li et al. (2024); Yang et al. (2024); Wu et al. (2024b); we add that *agentic behavior itself* — not only reasoning accuracy — is brittle to surface form, and that this brittleness is locatable upstream of retrieval.

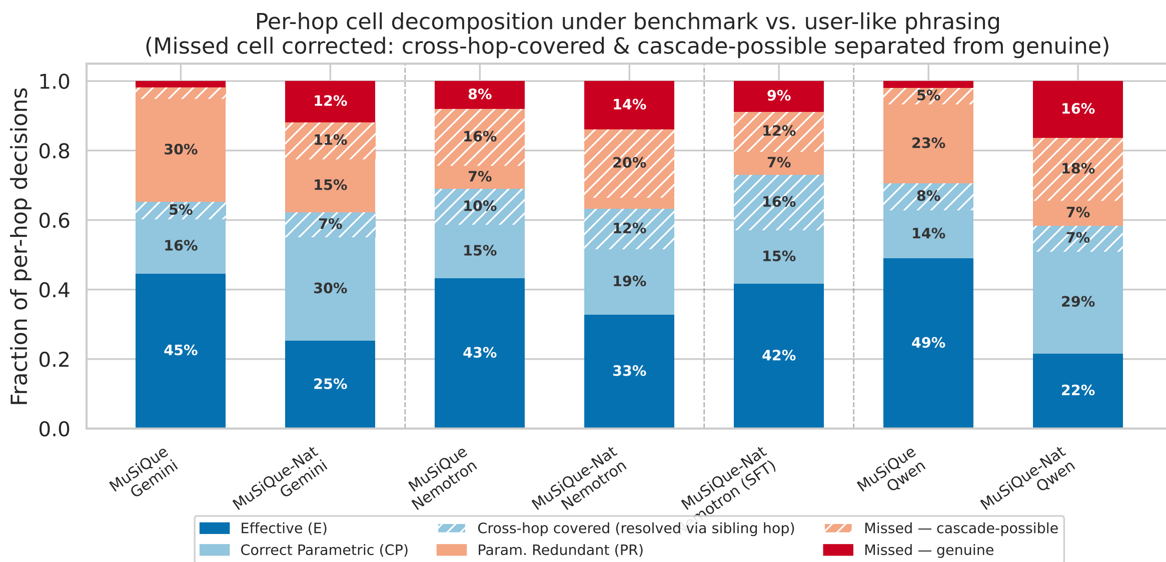


Figure 1: Per-hop behavior collapsed into four cells per (model, condition): green cells are correct decisions (Effective search E; Correct Parametric CP), red cells are errors (Missed M; Parametrically Redundant PR). For §4.1 the relevant columns are the three untuned models in both benchmark (MuSiQue) and user-like (MuSiQue-Natural) conditions: across all three models the user-like rewrite contracts the green E cell and expands the red M cell. The fourth column for Nemotron — the LoRA-fine-tuned variant — is the subject of §4.5.

3 Diagnostic Protocol

We use a single hop-level protocol as the diagnostic instrument throughout the paper. For each multi-hop question q with gold hop decomposition $\{h_j\}$, we (1) sample the agent M five times in closed-book mode and compute the per-hop semantic entropy $\mathcal{E}_j$ via LLM-judge semantic clustering (Farquhar et al., 2024); (2) run M on the full question with web search enabled, recording the trace; (3) use an LLM judge to attribute every emitted search query to a specific hop j. The cross-product of (certain $\mathcal{E}_j$=0, uncertain $\mathcal{E}_j$>0) × (searched, not searched), refined with two trace-level flags (sequential redundancy, cross-hop coverage), partitions every per-hop decision into six interpretable cells. The same protocol applies unchanged across benchmark phrasing and user-style paraphrase, which is why we use it. Full prompts, judge choices, and category definitions are deferred to the appendix; the only fact a reader needs at this point is that the protocol returns, per (model, question), the search volume, the per-hop search decisions, and the resulting six-cell behavior counts.

4 Experiments and Analysis

We instantiate the protocol on three production-grade agentic systems — Gemini 3.1 Pro, Qwen 3.5 (122B), and Nemotron Nano (30B) — and 600 staleness-filtered MuSiQue (Trivedi et al., 2022) questions in two conditions (benchmark phrasing and a natural, user-styled rewrite preserving the same information need), plus a curated slice of real user queries (ShareChat) for which only coarse search-volume signal is available.

4.1 The gap is real and large (behavior)

Two findings, drawn from a single experiment.

(F1) Search volume collapses under user-like phrasing. Mean searches per question drop from 15.20→4.93 for Gemini (3.1×), 5.74→3.13 for Nemotron (1.8×), and 5.60→1.33 for Qwen (4.2×); all paired differences are highly significant ($p < 10^{-21}$). The agent does not just call the tool less — it calls it much less.

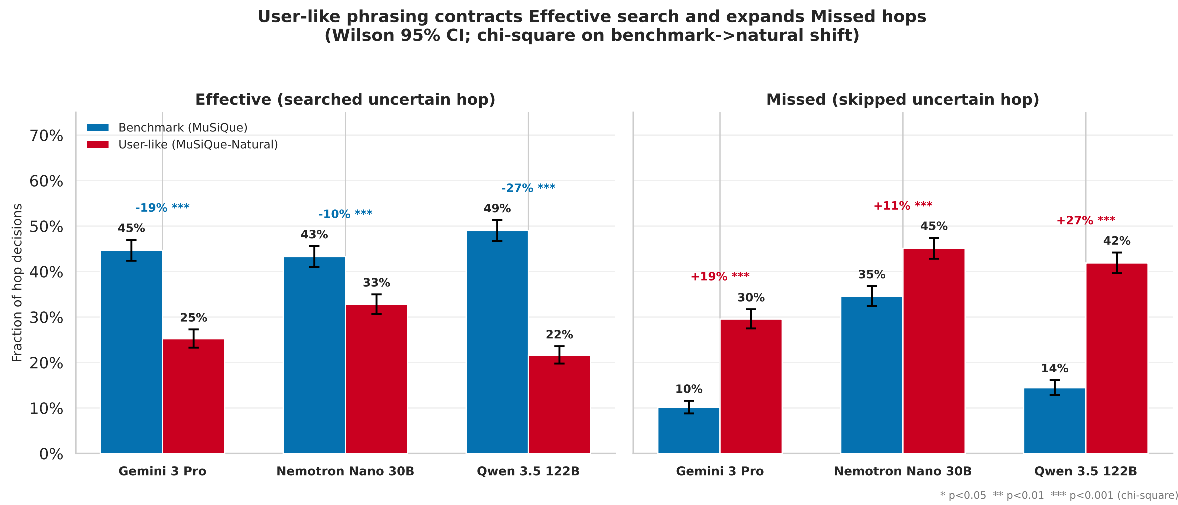


Figure 2: The two load-bearing cells of Figure 1 — Effective (E) and Missed (M) — under benchmark (MuSiQue) vs. user-like (MuSiQue-Natural) phrasing, with Wilson 95% confidence intervals. The benchmark→user-like shift contracts E and expands M for every model; each within-model shift is χ^2-significant ($p < 10^{-9}$). This is the significance-and-uncertainty view of finding F2.

(F2) The surviving searches are mis-targeted. The dominant cell-level change in Figure 1 is $E \to M$: the agent stops searching on the very hops it itself flagged as uncertain in closed-book sampling. The Missed cell grows by 10.6–27.4 percentage points depending on the model (Effective contracts by a matching 10.5–27.4 pp), and the surviving searches are reallocated *away from* the uncertain hops the same model had identified. Figure 2 isolates these two load-bearing cells with Wilson 95% confidence intervals; every per-model E contraction and M expansion is χ^2-significant at $p < 10^{-9}$.

4.2 The Missed cell is costly, not benign

A reader could object that M is not really an error: end-task accuracy in fact *rises* under the user-like rewrite (e.g. +10 pp for Gemini), so perhaps the agent is sensibly economizing on hops it can resolve by reasoning. We rule this out at two levels. We use the per-hop *in-trace* answer recovered by the commitment-locus judge (§4.3) and, to be conservative, define a hop as *Truly missed* only if it is uncertain, unsearched, *and* its gold answer does not appear in any other hop's retrieved snippets — so hops the agent resolved by cross-hop retrieval are excluded from the count.

(F2a) Per-hop: skipped uncertain hops are answered worse. On uncertain hops, the ones the agent skips are answered correctly far less often *in the actual trace* than the ones it searches: in the user-like condition, 25.8% vs. 32.6% for Gemini, 20.4% vs. 48.0% for Nemotron, and 20.9% vs. 51.9% for Qwen (Figure 3; five of six model×condition gaps are significant by a two-proportion test, the lone exception being Gemini under benchmark phrasing, where it skips only hops it already knows).

(F2b) End-to-end: a missed hop lowers final accuracy at fixed difficulty. Questions containing ≥ 1 Truly-missed hop are answered correctly far less often, and this survives stratification by hop count: a Mantel–Haenszel test pooling over the 2/3/4-hop strata gives a common odds ratio of 0.24 ($p < 10^{-12}$) under benchmark phrasing and 0.46 ($p < 10^{-13}$) under the user-like rewrite. The aggregate accuracy gain is therefore a coarse-question-difficulty artifact; *conditional on difficulty, the Missed cell costs correctness.* This is what licenses treating M as an error rather than as economy.

4.3 The mechanism: pre-search commitment, not cascade

Two probes turn the behavioral finding into a mechanistic one by ruling out the two natural alternative explanations for the missed-cell growth — "the model delayed the search but still got the answer from later context" and "the missing hops are downstream cascade failure from an upstream wrong hop."

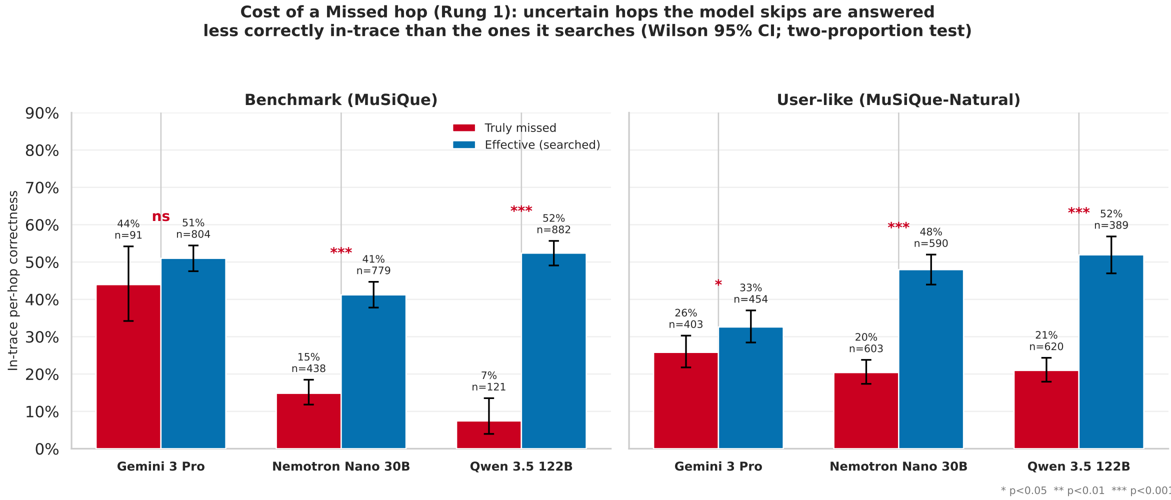


Figure 3: Cost of a Missed hop (per-hop view). On parametrically uncertain hops, in-trace per-hop correctness is lower for *Truly missed* hops (uncertain, unsearched, not cross-hop-covered) than for *Effective* (searched) hops, in every model×condition except Gemini-benchmark. Wilson 95% CIs; two-proportion test stars. The end-to-end, difficulty-stratified version (per-stratum accuracy with Mantel–Haenszel odds ratios) is provided in the supplementary material.

Read together (Figures 4, 5), the two probes establish a single mechanism: under user-like phrasing the agent fixes a per-hop answer *before* issuing search for that hop, and the resulting Missed cell is genuine missing rather than downstream cascade from upstream failure. This is the load-bearing claim of the paper.

4.4 The gap, in the wild (ShareChat)

A natural worry about §§4.1–4.3 is that the gap is an artifact of the rewriting procedure itself. We address this with a curated slice of real user queries from ShareChat, run with the same three agents.

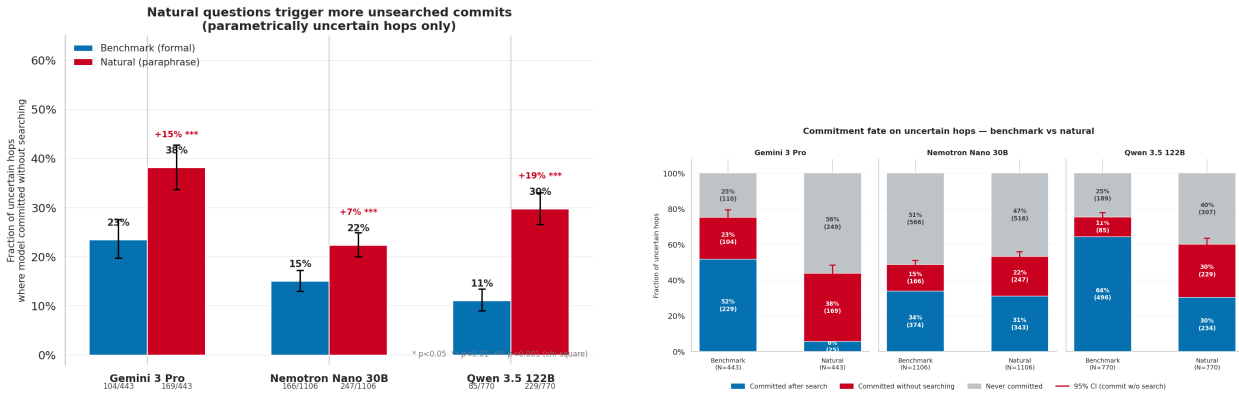
We can only report the coarse signal here, because ShareChat lacks the gold hop decomposition needed for the full per-hop diagnostic; obtaining sub-hop labels and replicating §4.3's commitment-locus and cascade analyses on real user queries is the natural next step.

4.5 A first intervention: closing the gap on Nemotron

If pre-search commitment under user-like phrasing is the mechanism (§4.3), then a training-time intervention that targets it should be visible as a contraction of the Missed cell in the user-like condition. We fine-tune Nemotron Nano 30B with LoRA (rank 64, α=128, 3 epochs, target modules `q,k,v,o_proj`, `lm_head`) on a curated SFT mix drawn from the model's own rollouts: (i) on-policy traces where Nemotron answered correctly *and* searched every uncertain hop, (ii) no-search-enrichment traces from cases where the model was simultaneously certain and correct, and (iii) phrasing augmentation that pairs each formal MuSiQue question with its natural-language paraphrase so the same information need is seen under both surface forms.

The result is the fourth column of Figure 1 (Nemotron with SFT, MuSiQue-Natural). The Missed cell drops from 45.1% to 36.2% ($\Delta = -8.9$ pp), and the Effective cell rises from 32.8% to 41.7% ($\Delta = +8.9$ pp): the intervention reroutes nearly nine percentage points of search decisions from "missed an uncertain hop" to "searched an uncertain hop." The trade-off is real and worth flagging: Parametrically Redundant rises 3.3%→6.7% (+3.4 pp) and mean searches per question rise 2.49→4.38 (+76%).

Behavioral, not yet end-to-end: the SFT under-targets cost. End-task accuracy moves only 36.0%→36.8% (+0.8 pp, not significant), so the behavioral fix does not (yet) buy correctness. The targeting diagnostic of §4.2 explains why. Of the base model's Truly-missed hops, 79.6% are high-cost (answered wrong in-trace when missed) — exactly the hops a search could rescue. Yet the SFT converts (now searches) only 31.5% of those high-cost hops, while converting 44.7% of the low-cost ones (right anyway despite missing) —



(a) Pre-search commitment rate.

(b) Commitment-fate breakdown.

Figure 4: Commitment-locus probe (parametrically uncertain hops only, $\mathcal{E}_j > 0$). **(a)** Per (model, variant) fraction of uncertain-hop decisions where the agent has committed to a specific per-hop answer *before* any search query attributable to that hop is issued. Under user-like phrasing this rate rises for all three models (23.5%→38.1% for Gemini; 11.0%→29.7% for Qwen; 15.0%→22.3% for Nemotron; all χ^2-significant). **(b)** The same per-hop decisions are split by what happens to the committed answer. The orange "committed without ever searching" band expands under the rewrite for every model. **What these support:** the locus of the per-hop answer moves *upstream* of retrieval — the agent is not delaying retrieval, it is substituting parametric commitment for it, and doing so on hops the same model had flagged as uncertain in closed-book sampling.

a conversion bias *toward* the hops that did not need fixing (two-proportion $p = 0.006$). Because high-cost hops dominate the pool, most conversions are still high-cost in absolute terms (73%), but the recipe leaves two-thirds of the rescuable hops uncovered and spends a disproportionate share of its new searches where they cannot help. Combined with the fact that searching a previously-missed hop only lifts its in-trace correctness to $\sim 48\%$ (§4.2), the small accuracy yield is expected. We therefore treat this as a proof of concept that the gap is *behaviorally* closable by fine-tuning, and as direct evidence that the next intervention must be cost-aware — reward searching the hops the model would otherwise get wrong, not uncertain hops indiscriminately — rather than as a finished solution.

5 Discussion

Why the gap exists. The behavioral and mechanistic findings are mutually consistent with a single picture: agentic search policy is anchored to surface features of the prompt (explicit X-of-Y phrasing, named bridge entities, enumeration syntax) more than to the agent's internal representation of what it does and does not know. The benchmark phrasing exposes those features explicitly; the user-like phrasing strips them. The agent *can still answer the closed-book parts equally well* — semantic entropy on the same facts is unchanged — but loses the cue that previously triggered the tool call, and instead commits to a parametric answer *before* issuing search (§4.3). This fits prior work on the knowing-doing gap (Cheng et al., 2026) and on function-calling collapse under rephrasing (Rabinovich et al., 2025): the agent recognizes a fact gap internally and still does not act.

Why it matters. Agentic LLM systems are increasingly evaluated using benchmark-style multi-hop questions, and the resulting tool-use claims — "model M searches when it should and skips when it shouldn't" — are treated as portable across deployment surfaces. Our results show they are not portable. A benchmark-derived behavior profile is, at best, an upper bound on the same agent's behavior on realistic user queries; for the three models we measured, it is a misleading upper bound. Any agentic-LLM safety claim, cost claim, or trust claim grounded in benchmark tool-use behavior should be treated as load-bearing only after a distribution-shift audit that includes user-like surface forms.

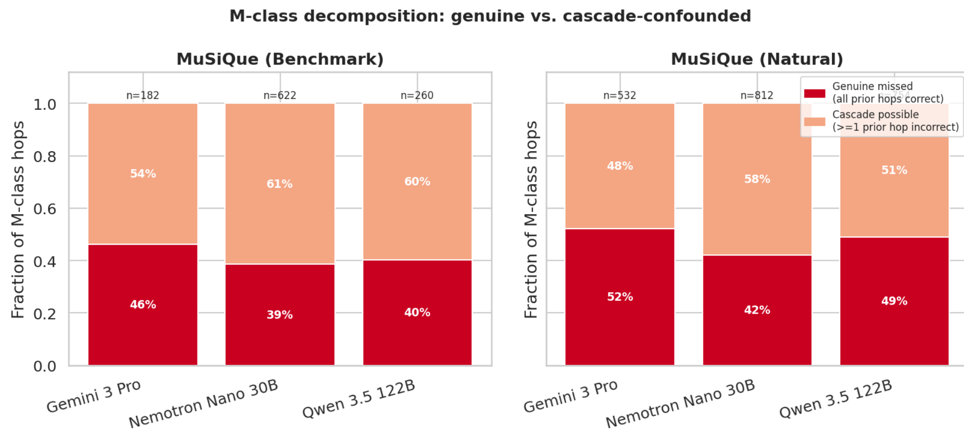


Figure 5: Decomposition of the Missed cell into a "genuine missed" component (hops where every upstream hop was correctly answered, so the missing search is not explainable by upstream failure) and a "cascade-possible" component (hops where some upstream hop was wrong, so the missing could in principle be downstream collateral damage). **What this supports:** the dark "genuine missed" component grows under user-like phrasing for every model, ruling out cascade-driven explanations. The Missed-cell expansion is real missing, not artifact.

What closes the gap (partially). §4.5 demonstrates that surface-form-augmented SFT on on-policy rollouts filtered for uncertainty-aware tool use is enough to move nearly nine percentage points of per-hop decisions out of the Missed cell on the user-like distribution of one model. This is a concrete training-time fix, not a hypothetical. It is also Pareto-bounded: the cost is a 76% rise in mean searches per question and a small rise in Parametrically Redundant. The most plausible next interventions are reward shaping that explicitly penalizes pre-search commitment on uncertain hops, and mandatory-retrieval gates as a deployment-side fallback when the model's surface-form-anchored policy cannot be trusted. The diagnostic instrument introduced here makes "did this intervention actually fix the gap" a directly measurable question, not a downstream-only one.

6 Conclusion

We asked whether the agentic search behavior of frontier LLMs transfers from benchmark-style multi-hop questions to user-like questions. Using a hop-level diagnostic protocol on three production-grade agentic systems and 600 MuSiQue questions, we found four things. **(1)** The gap is real and large: search volume collapses 1.8–4.2× under user-like rewriting and the surviving searches are mis-targeted away from the agent's own uncertain hops. **(2)** The mechanism is upstream of retrieval — pre-search commitment rises sharply under the rewrite and the missed-cell expansion is genuine missing rather than downstream cascade failure. **(3)** The direction of effect reproduces on real user queries from a curated ShareChat slice; this is not an artifact of paraphrase rewriting. **(4)** The gap is closable, partially: an LoRA fine-tune of Nemotron on phrasing-augmented and uncertainty-aware on-policy rollouts cuts the Missed cell by 8.9 pp on the user-like distribution at the cost of a 76% rise in mean searches. The diagnostic instrument makes "did the intervention transfer too?" a directly measurable question, which is precisely what is needed to evaluate interventions on what they actually have to fix — behavior under the surface forms real users produce.

Limitations

- **Domain coverage.** Our mechanistic experiments use one open-domain multi-hop QA family (MuSiQue) and three frontier models; we do not yet measure the gap on code-agent, tool-orchestration, or scientific-research agent surfaces.

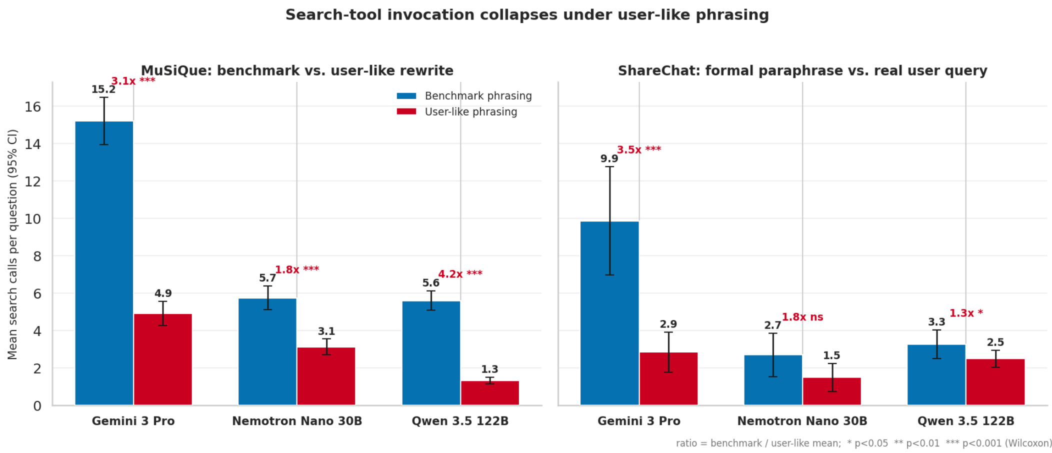


Figure 6: Search-tool invocation on real user queries from a curated ShareChat slice, alongside MuSiQue (benchmark) and MuSiQue-Natural (user-styled rewrite). The MuSiQue → MuSiQue-Natural drop is matched in direction by the benchmark → ShareChat drop across all three models. The gap observed under controlled paraphrasing is not a rewriting artifact; it shows up directly on real user queries.

- **ShareChat lacks hop labels.** On real user queries we can only report search-volume direction, not the per-hop mechanistic version of the same finding. Replicating the commitment-locus and cascade-decomposition analyses on user queries requires sub-hop annotation and is deferred to future work.
- **LLM-judge noise.** Per-hop attribution and semantic clustering depend on LLM judges; we follow Lee and Hockenmaier (2025) and Han et al. (2025) for validity, but residual noise inflates the category counts in the worst case.
- **Single-turn agent traces.** We do not yet measure how the gap compounds across multi-turn conversational agents, where each turn's surface-form choice may further shift the next turn's tool-use policy.
- **Single-model intervention.** The training-time fix in §4.5 is demonstrated on Nemotron Nano 30B only. Replicating the LoRA SFT recipe on Gemini and Qwen is straightforward in principle but not yet done.
- **Pareto frontier not characterized.** The SFT intervention trades coverage of uncertain hops against a 76% rise in mean searches per question. We do not yet know what the achievable coverage/over-search Pareto frontier looks like for this recipe, or whether a reward-shaping variant could dominate it.

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